INDUSTRIAL AUTOMATION ENGINEERING PORTFOLIO

Project 4 — Version 2

Industrial Temperature Control System

PLC / Simulation Portfolio Project

Version 2 — Structured Controls Upgrade

1. Project Overview

Project 4 Version 2 upgrades the original temperature-control project into a structured PLC-controlled heating system. I separated operating permission, Auto/Manual demand, heater request, startup timing, cooling control, temperature diagnostics, alarm handling, and final output control rather than allowing a temperature comparison to directly control the heater.

The final tested ladder uses Auto_Mode I:1/3 and Manual_Mode I:1/4, Start_PB I:1/0, Stop_PB I:1/1, Reset_PB I:1/2, EStop_OK I:1/5, and Overload_OK I:1/6. The process value is stored in N7:0. The final outputs are Heater_ON O:2/0, Cooling_Fan O:2/1, Lamp O:2/2, Fault_Lamp O:2/3, and Alarm_Horn O:2/4.

The program also includes a latched Auto_Heating_Request B3:0/0, non-latched Manual_Run_Request B3:0/1, Heater_Run_Request B3:0/2, System_Ready B3:0/3, Fault_Active B3:0/4, Mode_Conflict B3:0/5, High_Temperature_Condition B3:0/6, Start_Permissive B3:0/7, and latched Sensor_Failure B3:1/0.

This project was developed and tested in a PLC training/simulation environment. The PLC monitoring shown for E-stop, overload, temperature, faults, and outputs is not presented as physical-machine commissioning or as a complete safety-rated heating system.

2. Engineering Objective

- Provide separate Auto and Manual operating modes.
- Detect simultaneous Auto/Manual selection through Mode_Conflict.
- Monitor E-stop and overload healthy conditions.
- Detect out-of-range temperature values and latch Sensor_Failure.
- Create a centralized Start_Permissive and System_Ready state.
- Generate a latched automatic heating request from the implemented temperature comparison.
- Provide a separate non-latched Manual heating request.
- Combine both requests into Heater_Run_Request.
- Apply a programmed three-second startup delay before heater operation.
- Provide independent Cooling Fan control.
- Detect a high-temperature condition and delay its audible alarm response by five seconds.
- Provide visual and audible fault indication.
- Inhibit the final Heater_ON output for Fault_Active or High_Temperature_Condition.

3. Functional Requirements

- Mode_Conflict B3:0/5 shall become true when Auto_Mode I:1/3 and Manual_Mode I:1/4 are selected simultaneously.
- Fault_Active B3:0/4 shall latch from the implemented mode-conflict, E-stop, overload, or Sensor_Failure conditions.
- Reset_PB I:1/2 shall unlatch Fault_Active only while EStop_OK and Overload_OK are healthy and Mode_Conflict is false.
- Sensor_Failure B3:1/0 shall be reset only when the temperature is within the implemented valid range during the Reset sequence.
- Start_Permissive B3:0/7 shall require Fault_Active false, EStop_OK true, Overload_OK true, and Mode_Conflict false.
- System_Ready B3:0/3 shall require Start_Permissive and either Auto or Manual mode.
- Auto_Heating_Request B3:0/0 shall latch in Auto mode when System_Ready is true and N7:0 is less than N7:1.
- Auto_Heating_Request shall unlatch when N7:0 is greater than or equal to N7:1, Stop_PB is active, Fault_Active is true, or Auto_Mode is removed.
- Manual_Run_Request B3:0/1 shall energize while Manual_Mode, Start_PB, and System_Ready are true.

- Heater_Run_Request B3:0/2 shall energize from either Auto_Heating_Request or Manual_Run_Request while System_Ready remains true.
- T4:0 shall use a 0.01-second time base and preset of 300, providing a programmed three-second delay.
- Heater_ON O:2/0 shall be controlled by the implemented heater command logic and final fault/high-temperature interlocks.
- Cooling_Fan O:2/1 shall operate from System_Ready, the N7:1 > N7:2 comparison, and XIO Sensor_Failure.
- Lamp O:2/2 shall follow Heater_ON O:2/0.
- High_Temperature_Condition B3:0/6 shall energize when N7:0 is greater than 100.
- T4:1 shall use a 0.01-second time base and preset of 500, providing a five-second alarm delay.
- Sensor_Failure B3:1/0 shall latch when N7:0 is less than 0 or greater than 200.
- Fault_Lamp O:2/3 shall follow Fault_Active.
- Alarm_Horn O:2/4 shall energize from Fault_Active or T4:1/DN.
- The final Heater_ON rung shall require T4:0/DN and Heater_Run_Request while Fault_Active and High_Temperature_Condition are false.

4. System Components

Component	Address / Identification	Function
Start Push Button	I:1/0	Operator Start command
Stop Push Button	I:1/1	Automatic heating request stop condition
Reset Push Button	I:1/2	Fault and sensor-failure recovery command
Auto Mode Selector	I:1/3	Automatic mode selection
Manual Mode Selector	I:1/4	Manual mode selection
E-stop Healthy	I:1/5	E-stop circuit healthy feedback
Overload Healthy	I:1/6	Overload healthy feedback
Temperature Value	N7:0	Process value
Auto Heating Setpoint	N7:1	Automatic heating comparison reference
Cooling Reference	N7:2	Cooling comparison reference
Heater ON	O:2/0	Heater output command
Cooling Fan	O:2/1	Cooling output command
Lamp	O:2/2	Heater-command indication
Fault Lamp	O:2/3	Visual fault indication
Alarm Horn	O:2/4	Audible alarm indication
Delay Timer	T4:0	Three-second heater startup delay
Alarm Delay	T4:1	Five-second high-temperature alarm delay

5. PLC Data and I/O Assignment

Function	Address	Type
Start_PB	I:1/0	Digital Input
Stop_PB	I:1/1	Digital Input
Reset_PB	I:1/2	Digital Input
Auto_Mode	I:1/3	Digital Input
Manual_Mode	I:1/4	Digital Input
EStop_OK	I:1/5	Digital Input
Overload_OK	I:1/6	Digital Input
Heater_ON	O:2/0	Digital Output
Cooling_Fan	O:2/1	Digital Output

Lamp	O:2/2	Digital Output
Fault_Lamp	O:2/3	Digital Output
Alarm_Horn	O:2/4	Digital Output
Auto_Heating_Request	B3:0/0	Internal Bit
Manual_Run_Request	B3:0/1	Internal Bit
Heater_Run_Request	B3:0/2	Internal Bit
System_Ready	B3:0/3	Internal Bit
Fault_Active	B3:0/4	Latched Internal Bit
Mode_Conflict	B3:0/5	Internal Bit
High_Temperature_Condition	B3:0/6	Internal Bit
Start_Permissive	B3:0/7	Internal Bit
Sensor_Failure	B3:1/0	Latched Internal Bit
Delay_Timer	T4:0	TON
Alarm_Delay	T4:1	TON
Temperature Value	N7:0	Integer
Heating Setpoint	N7:1	Integer
Cooling Reference	N7:2	Integer

6. Control Strategy

I organized the control path as Operating Permission → Heating Demand → Common Heater Request → Startup Timing → Final Heater Output. Auto mode uses a latched heating request based on N7:0 < N7:1; Manual mode uses a non-latched request based on Manual_Mode, Start_PB, and System_Ready.

Temperature diagnostics are separated from normal demand. High_Temperature_Condition B3:0/6 is generated when N7:0 > 100 and drives Alarm_Delay T4:1. Sensor_Failure B3:1/0 is latched when N7:0 is outside the implemented 0–200 valid range. The final Heater_ON rung rechecks Fault_Active and High_Temperature_Condition before O:2/0 can remain energized.

7. Engineering Design Decisions

Design Decision	Engineering Purpose
Separate Auto and Manual modes	Provide distinct automatic and operator-controlled behavior
Latched Auto_Heating_Request	Maintain automatic demand after the comparison first becomes true
Mode Conflict bit	Detect invalid simultaneous selector operation
Start Permissive / System Ready	Separate machine health from actual heating demand
Common Heater Run Request	Share downstream timing/output logic between modes
T4:0 startup delay	Delay physical heater command after demand is accepted
Independent Cooling Fan logic	Allow cooling to respond to its own comparison and sensor-validity condition
High Temperature Condition	Create an explicit overtemperature state
T4:1 alarm delay	Require persistent high temperature for the delayed alarm path
Latched Sensor Failure	Preserve an out-of-range sensor event until a valid reset occurs
Final heater interlocks	Recheck fault and high-temperature state at output level

8. Ladder Logic Description

8.1 Rung 0000 — Mode Conflict Detection

```
|----[ XIC I:1/3 ]----[ XIC I:1/4 ]----------------------------( OTE B3:0/5 )----|
      Auto_Mode       Manual_Mode                               Mode_Conflict
```

Rung Description: Both Auto_Mode I:1/3 and Manual_Mode I:1/4 must be true for Mode_Conflict B3:0/5 to energize. Because an OTE is used, the bit follows the present selector condition.

Engineering Purpose: I included this rung to identify invalid simultaneous mode selection.

8.2 Rung 0001 — Master Fault Latch

```
     +----[ XIC B3:0/5 ]-----------------------------------------------+
     |     Mode_Conflict                                                |
     +----[ XIO I:1/5 ]------------------------------------------------+
     |     EStop_OK                                                     |
|----+----[ XIO I:1/6 ]------------------------------------------------+----( OTL B3:0/4 )----|
     |     Overload_OK                                                  |      Fault_Active
     +----[ XIC B3:1/0 ]-----------------------------------------------+
           Sensor_Failure
```

Rung Description: Any implemented fault source can latch Fault_Active B3:0/4. The E-stop and overload inputs are healthy-state signals, so XIO detects loss of the healthy condition.

Engineering Purpose: The latched master fault prevents a transient abnormal condition from disappearing without operator recovery.

8.3 Rung 0002 — Fault and Sensor Failure Reset

```
|----[ XIC I:1/2 ]--[ XIC I:1/5 ]--[ XIC I:1/6 ]--[ XIO B3:0/5 ]----+----( OTU B3:0/4 )----|
      Reset_PB      EStop_OK       Overload_OK   Mode_Conflict     |      Fault_Active
                                                                   |
                                                                   +----[ GEQ N7:0 0 ]--[ LES N7:0 201 ]--( OTU B3:1/0 )----|
                                                                        Valid temperature range              Sensor_Failure
```

Rung Description: Reset_PB must be true, EStop_OK and Overload_OK must be healthy, and Mode_Conflict must be false. Fault_Active is unlatched directly. Sensor_Failure is unlatched only when the temperature value is within the implemented valid range N7:0 >= 0 and N7:0 < 201.

Engineering Purpose: I used conditional recovery so the master fault and sensor diagnostic cannot be cleared without the required healthy conditions.

8.4 Rung 0003 — Start Permissive

```
|----[ XIO B3:0/4 ]----[ XIC I:1/5 ]----[ XIC I:1/6 ]----[ XIO B3:0/5 ]----( OTE B3:0/7 )----|
     Fault_Active       EStop_OK         Overload_OK      Mode_Conflict      Start_Permissive
```

Rung Description: Start_Permissive B3:0/7 is true only when Fault_Active is false, EStop_OK and Overload_OK are true, and no Mode_Conflict exists.

Engineering Purpose: This gives the rest of the program one centralized operating-permission status.

8.5 Rung 0004 — System Ready

```
                                     +----[ XIC I:1/3 ]----+
                                     |     Auto_Mode       |
|----[ XIC B3:0/7 ]------------------+                     +----( OTE B3:0/3 )----|
      Start_Permissive               |                     |      System_Ready
                                     +----[ XIC I:1/4 ]----+
                                           Manual_Mode
```

Rung Description: System_Ready B3:0/3 requires Start_Permissive and either Auto_Mode I:1/3 or Manual_Mode I:1/4.

Engineering Purpose: This separates general machine permission from an actual heating request.

8.6 Rung 0005 — Automatic Heating Request Latch

```
|----[ XIC I:1/3 ]----[ XIC B3:0/3 ]----[ LES N7:0 N7:1 ]------------( OTL B3:0/0 )----|
       Auto_Mode       System_Ready      Temperature < Setpoint       Auto_Heating_Request
```

Rung Description: In Auto mode, System_Ready and the LES comparison N7:0 < N7:1 latch Auto_Heating_Request B3:0/0.

Engineering Purpose: The latched internal request separates process demand from the physical heater output.

8.7 Rung 0006 — Automatic Heating Request Unlatch

```
     +----[ GEQ N7:0 N7:1 ]---------------------------------------------+
     |     Temperature >= Setpoint                                      |
     +----[ XIC I:1/1 ]-------------------------------------------------+
     |     Stop_PB                                                      |
|----+----[ XIC B3:0/4 ]------------------------------------------------+----( OTU B3:0/0 )----|
     |     Fault_Active                                                 |      Auto_Heating_Request
     +----[ XIO I:1/3 ]-------------------------------------------------+
           Auto_Mode
```

Rung Description: Auto_Heating_Request B3:0/0 is unlatched when the process reaches/exceeds N7:1, Stop_PB is active, Fault_Active is true, or Auto_Mode is removed.

Engineering Purpose: This rung defines all implemented automatic shutdown conditions in one location.

8.8 Rung 0007 — Manual Heating Request

```
|----[ XIC I:1/4 ]----[ XIC I:1/0 ]----[ XIC B3:0/3 ]----------------( OTE B3:0/1 )----|
      Manual_Mode      Start_PB         System_Ready                  Manual_Run_Request
```

Rung Description: Manual_Run_Request B3:0/1 is true only while Manual_Mode, Start_PB, and System_Ready are all true.

Engineering Purpose: The non-latched Manual request keeps operation dependent on the continuing operator command.

8.9 Rung 0008 — Common Heater Run Request

```
     +----[ XIC B3:0/0 ]-------+
     |    Auto_Heating_Request |
|----+                         +----[ XIC B3:0/3 ]----------------( OTE B3:0/2 )----|
     |                         |     System_Ready                  Heater_Run_Request
     +----[ XIC B3:0/1 ]-------+
          Manual_Run_Request
```

Rung Description: Either mode-specific request can energize Heater_Run_Request B3:0/2 while System_Ready remains true.

Engineering Purpose: This allows Auto and Manual modes to share the same startup and heater-control logic.

8.10 Rung 0009 — Heater Startup Delay

```
|----[ XIC B3:0/2 ]----------------------------------------[ TON T4:0 ]----|
       Heater_Run_Request                                  Delay_Timer  Time Base: 0.01 s  Preset: 300
```

Rung Description: T4:0 accumulates while Heater_Run_Request is true. With a 0.01-second time base and preset of 300, the programmed delay is three seconds.

Engineering Purpose: The timer prevents immediate heater energization after demand is accepted.

8.11 Rung 0010 — Heater Command

```
|----[ XIC T4:0/DN ]-------[ XIC B3:0/2 ]--------------------------( OTE O:2/0 )----|
       Delay_Timer_DN      Heater_Run_Request                        Heater_ON
```

Rung Description: After the startup delay reaches Done, Heater_ON O:2/0 energizes while Heater_Run_Request remains true.

Engineering Purpose: The request is rechecked after timing before the physical output is commanded.

8.12 Rung 0011 — Cooling Fan Control

```
|----[ XIC B3:0/3 ]----[ GRT N7:1 N7:2 ]------------[ XIO B3:1/0 ]------------( OTE O:2/1 )----|
      System_Ready      HeatingRef > CoolingRef     Sensor_Failure             Cooling_Fan
```

Rung Description: Cooling_Fan O:2/1 energizes when System_Ready is true, N7:1 > N7:2, and Sensor_Failure is false.

Engineering Purpose: I kept cooling control on its own logic path and inhibited it for an active sensor diagnostic.

8.13 Rung 0012 — Heater Lamp

```
|----[ XIC O:2/0 ]---------------------------------------------------( OTE O:2/2 )----|
       Heater_ON                                                           Lamp
```

Rung Description: Lamp O:2/2 follows the Heater_ON O:2/0 command.

Engineering Purpose: This provides direct indication of the PLC heater command.

8.14 Rung 0013 — High Temperature Condition

```
|----[ GRT N7:0 100 ]----------------------------------------------( OTE B3:0/6 )----|
       Temperature > 100                                                     High_Temperature_Condition
```

Rung Description: High_Temperature_Condition B3:0/6 becomes true when N7:0 is greater than 100.

Engineering Purpose: This creates a dedicated overtemperature state independent from normal heating demand.

8.15 Rung 0014 — High Temperature Alarm Delay

```
|----[ XIC B3:0/6 ]----------------------------------------[ TON T4:1 ]----|
       High_Temperature_Condition                          Alarm_Delay  Time Base: 0.01 s  Preset: 500
```

Rung Description: T4:1 accumulates while High_Temperature_Condition is true. The 0.01-second time base and preset 500 create a five-second delay.

Engineering Purpose: The delay requires the high-temperature condition to persist before the delayed alarm path becomes true.

8.16 Rung 0015 — Sensor Failure Latch

```
     +----[ LES N7:0 0 ]-----------------------------------------------+
     |     Temperature < 0                                             |
|----+                                                                 +----( OTL B3:1/0 )----|
     |                                                                 |      Sensor_Failure
     +----[ GRT N7:0 200 ]---------------------------------------------+
           Temperature > 200
```

Rung Description: Sensor_Failure B3:1/0 latches when N7:0 is below 0 or above 200.

Engineering Purpose: The latched diagnostic preserves an out-of-range temperature event until a valid Reset condition is satisfied.

8.17 Rung 0016 — Fault Lamp

```
|----[ XIC B3:0/4 ]--------------------------------------------------( OTE O:2/3 )----|
       Fault_Active                                                    Fault_Lamp
```

Rung Description: Fault_Lamp O:2/3 follows the latched Fault_Active B3:0/4 state.

Engineering Purpose: This provides a dedicated visual fault indication.

8.18 Rung 0017 — Alarm Horn

```
     +----[ XIC B3:0/4 ]----+
     |     Fault_Active     |
|----+                      +-------------------------------------------( OTE O:2/4 )----|
     |                      |                                             Alarm_Horn
     +----[ XIC T4:1/DN ]---+
           Alarm_Delay_DN
```

Rung Description: Alarm_Horn O:2/4 energizes from either Fault_Active or the completed Alarm_Delay.

Engineering Purpose: This combines immediate master-fault warning with delayed high-temperature warning.

8.19 Rung 0018 — Final Heater Output Interlock

```
|----[ XIC T4:0/DN ]----[ XIC B3:0/2 ]----[ XIO B3:0/4 ]----[ XIO B3:0/6 ]---------( OTE O:2/0 )----|
     Delay_Timer_DN   Heater_Run_Request   Fault_Active     High_Temp_Condition      Heater_ON
```

Rung Description: The final Heater_ON O:2/0 rung requires Delay Timer Done and Heater_Run_Request while Fault_Active and High_Temperature_Condition are false.

Engineering Purpose: This provides a final output-level interlock against fault and overtemperature conditions.

8.20 Rung 0019 — END

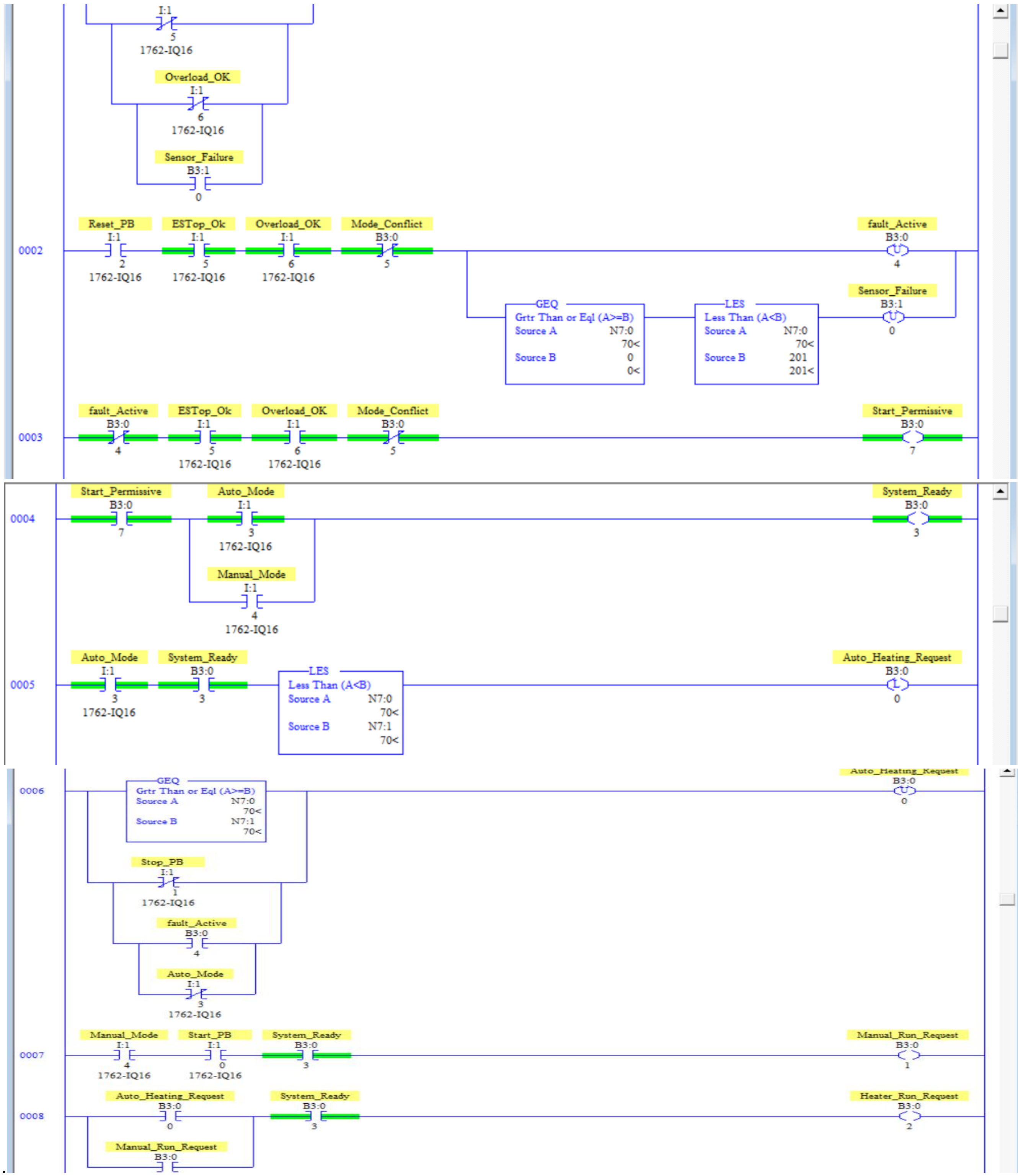
```
|------------------------------------------------------------------------------------------( END )----|
```

Rung Description: END marks the end of the main ladder program.

Engineering Purpose: This closes the PLC program file.

8.21 Actual RSLogix Implementation

The following images are the final Project 4 Version 2 ladder screenshots and are the technical source of truth for this portfolio. The typed rung representations above include the screenshot-confirmed PLC addresses and symbolic names. If any visual ambiguity remains, the final RSLogix screenshot governs.



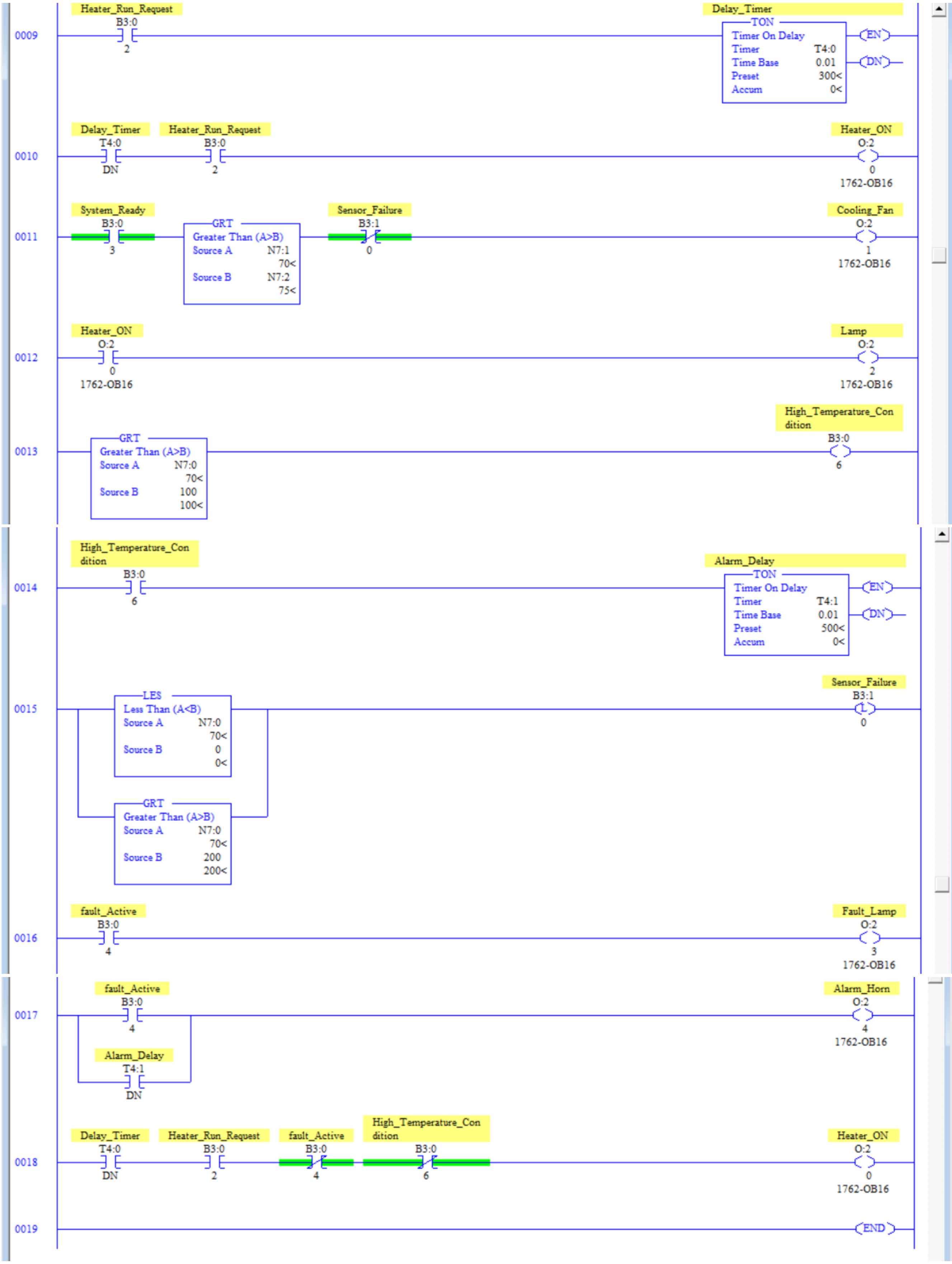


Figure P4V2.1 — Final RSLogix ladder screenshot.

9. Sequence of Operation

Step 1 — The PLC evaluates Mode_Conflict, EStop_OK, Overload_OK, Sensor_Failure, and Fault_Active.

Step 2 — Healthy conditions establish Start_Permissive and then System_Ready after Auto or Manual mode is selected.

Step 3 — In Auto mode, N7:0 < N7:1 latches Auto_Heating_Request.

Step 4 — The automatic request is removed when N7:0 >= N7:1, Stop_PB is active, Fault_Active is true, or Auto_Mode is removed.

Step 5 — In Manual mode, Manual_Mode + Start_PB + System_Ready generate Manual_Run_Request.

Step 6 — Either mode-specific request can establish Heater_Run_Request while System_Ready remains true.

Step 7 — Heater_Run_Request starts T4:0; after three seconds T4:0/DN becomes true.

Step 8 — Heater_ON O:2/0 is commanded when the timer and request conditions are satisfied.

Step 9 — Cooling_Fan O:2/1 is controlled by System_Ready, the N7:1 > N7:2 comparison, and Sensor_Failure status.

Step 10 — High_Temperature_Condition B3:0/6 becomes true when N7:0 > 100.

Step 11 — A persistent high-temperature condition starts T4:1 and completes the five-second delayed alarm path.

Step 12 — Sensor_Failure B3:1/0 latches if N7:0 < 0 or N7:0 > 200.

Step 13 — Fault_Lamp O:2/3 follows Fault_Active; Alarm_Horn O:2/4 follows Fault_Active or T4:1/DN.

Step 14 — The final Heater_ON rung inhibits O:2/0 for Fault_Active or High_Temperature_Condition.

10. Testing & Validation

Test	Validation Condition	Expected / Verified Response	Status
T-01	One valid mode; healthy conditions	Start_Permissive and System_Ready establish	PASS
T-02	Auto and Manual selected together	Mode_Conflict and fault path respond	PASS
T-03	N7:0 < N7:1 in Auto mode	Auto_Heating_Request latches	PASS
T-04	N7:0 >= N7:1	Auto_Heating_Request unlatches	PASS
T-05	Manual mode + Start	Manual_Run_Request energizes while enabled	PASS
T-06	Valid heating request	Heater_Run_Request energizes	PASS
T-07	Heater_Run_Request maintained	T4:0 completes programmed 3 s delay	PASS
T-08	Delay complete + valid request	Heater_ON O:2/0 energizes	PASS
T-09	Cooling logic satisfied	Cooling_Fan O:2/1 responds	PASS
T-10	N7:0 > 100	High_Temperature_Condition B3:0/6 becomes true	PASS
T-11	High temperature maintained	T4:1 completes programmed 5 s delay	PASS
T-12	N7:0 < 0	Sensor_Failure B3:1/0 latches	PASS
T-13	N7:0 > 200	Sensor_Failure B3:1/0 latches	PASS
T-14	Valid Reset with N7:0 in range	Sensor_Failure and master fault recovery logic operate	PASS
T-15	Fault Active during heating demand	Final Heater_ON rung inhibits O:2/0	PASS
T-16	High Temperature Condition during demand	Final Heater_ON rung inhibits O:2/0	PASS

Validation is documented as PLC/simulation testing of the final ladder implementation. It is not physical-machine commissioning, thermal-process validation, or safety certification.

11. Results

Project 4 Version 2 successfully expanded the original comparison-based temperature project into a structured heating-control application. The final screenshot-confirmed architecture separates Auto/Manual demand, common heater request, startup timing, cooling control, high-temperature monitoring, latched sensor-failure diagnostics, alarms, and final heater output interlocks.

The final address map is now consistent throughout this document: Heater_ON O:2/0, Cooling_Fan O:2/1, Lamp O:2/2, Fault_Lamp O:2/3, Alarm_Horn O:2/4, High_Temperature_Condition B3:0/6, and Sensor_Failure B3:1/0.

12. Challenges & Troubleshooting

Symptom	Diagnostic Path
System Ready OFF	I:1/3 / I:1/4 → B3:0/5 → B3:0/4 → I:1/5 / I:1/6 → B3:0/7 → B3:0/3
Auto heat request missing	I:1/3 → B3:0/3 → LES N7:0 N7:1 → B3:0/0
Manual request missing	I:1/4 → I:1/0 → B3:0/3 → B3:0/1
T4:0 not timing	B3:0/2 → T4:0
Heater OFF after delay	T4:0/DN → B3:0/2 → B3:0/4 → B3:0/6 → O:2/0
Cooling Fan not operating	B3:0/3 → GRT N7:1 N7:2 → B3:1/0 → O:2/1
High-temperature alarm missing	GRT N7:0 100 → B3:0/6 → T4:1/DN → O:2/4
Sensor Failure active	N7:0 versus 0 and 200 limits → B3:1/0
Fault Lamp remains ON	B3:0/4 → O:2/3

13. Lessons Learned

This project strengthened my understanding of using process values inside a structured PLC architecture. I learned to separate temperature demand from physical equipment control, share downstream logic between Auto and Manual modes, and use latched range diagnostics for implausible sensor values. The final screenshot review also reinforced the importance of preserving exact PLC addresses in portfolio documentation rather than relying on labels alone.

14. Industrial Relevance

Temperature control is common in ovens, dryers, tanks, process heaters, HVAC-related systems, manufacturing equipment, and process industries. This project demonstrates relevant concepts including mode management, permissives, process-value comparisons, delayed equipment startup, independent cooling, overtemperature monitoring, sensor diagnostics, alarms, and final output interlocks. A real installation could additionally require analog scaling, transmitter diagnostics, PID control, feedback devices, independent overtemperature protection, and application-specific safety engineering.

15. Version 1 → Version 2 Engineering Progression

Version 1 Foundation	Version 2 Implementation	Engineering Benefit
Basic temperature comparison	Latched Auto heating demand	Separates process demand from output
Simple operating logic	Auto/Manual architecture	Distinct operating behaviors
Direct output control	Request → delay → final output	Clearer control path
Limited permissives	Start Permissive / System Ready	Centralized operating state
No mode conflict	Mode_Conflict B3:0/5	Invalid selection detection
Immediate heater response	T4:0 startup delay	Controlled startup

Basic temperature response	Cooling_Fan O:2/1 logic	Separate heating/cooling functions
Limited overtemperature handling	High_Temperature_Condition B3:0/6 + T4:1	Persistent-condition alarm path
No sensor validation	Latched Sensor_Failure B3:1/0	Detects and preserves implausible values
Limited fault indication	Fault_Lamp O:2/3 + Alarm_Horn O:2/4	Improved operator indication
Upstream control only	Final O:2/0 fault/high-temp interlocks	Output-level protection check

16. Project Summary

Project 4 Version 2 upgraded the original temperature-control project into a structured Industrial Temperature Control System. The corrected portfolio now uses the exact screenshot-confirmed I/O and internal addresses throughout the I/O tables, ladder representations, testing, troubleshooting, and engineering descriptions.

The final ladder separates operating permission, Auto/Manual demand, common Heater_Run_Request, T4:0 startup timing, Cooling_Fan control, High_Temperature_Condition, T4:1 delayed alarm logic, latched Sensor_Failure, Fault_Lamp, Alarm_Horn, and final Heater_ON interlocks. The project remains documented as a PLC/simulation implementation rather than physical-machine commissioning.